

How personal anticipations and perceived gratifications influence continuous use intention toward AI-driven chatbots? Moderating roles of perceived innovativeness and active involvement

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ABSTRACT

In recent years, AI-driven chatbots have evolved into a transformative technological paradigm, deeply embedded in domains including retail, customer service, healthcare, and education. This study endeavors to unveil the psychological and behavioral mechanisms by which AI-driven chatbots influence users' continuous use intention. Leveraging an integrated theoretical framework, the study employs a partial least squares-structural equation modeling (PLS-SEM) approach. A valid sample comprising 581 AI-driven chatbot users was amassed via an online research methodology, meticulously scrutinizing their continuous use intention. Obtained findings substantiate a positive association between personal anticipations and perceived innovativeness, alongside affirming the positive correlation between perceived gratifications and active involvement. Both perceived innovativeness and active involvement positively influence continuous use intention toward AI-driven chatbots. These findings not only advance the understanding of user continuous use intention toward AI-driven chatbots from a novel theoretical perspective but also provide practitioners with pragmatic approaches to strategically develop AI-driven chatbots and sustain user engagement in the contemporary human-AI interaction environment.

1. Introduction

The rapid advancement of artificial intelligence (AI) has accelerated the proliferation of AI-driven chatbots, which are increasingly embedded in domains such as healthcare, psychology, behavior, and marketing (Pathak et al., 2025; Peng et al., 2025; Shahzad et al., 2024; Zhu et al., 2022). Projections suggest that by 2025, AI chatbots may handle up to 95% of consumer interactions conducted online (Lu & Zhang, 2025). These AI-driven chatbots have seamlessly integrated into messaging applications, mobile interfaces, and websites, offering responsiveness to frequently commonplace inquiries, delivering always-on, personalized assistance, and facilitating the dissemination and customization of information (Cheng & Jiang, 2020; Chiang et al., 2022). Organizations increasingly leverage chatbots to improve operational efficiency by automating high-frequency, routine customer requests, which can reduce staffing demands and associated service costs (Wirtz et al., 2018). In addition, data derived from chatbot interactions provides firms with actionable insights into product and service

optimization and fine-tuning marketing decisions, which may strengthen customer satisfaction (Lu & Zhang, 2025).

Yet the benefits of ubiquity and personalization have not consistently resulted in users' continuous confidence and use intention. Growing concerns about algorithmic decision-making and the negative emotions associated with algorithmic aversion have eroded trust in chatbot-mediated services, thereby weakening users' anticipations and perceived gratifications and increasing their preference for human-provided assistance over automated interactions (Lu & Zhang, 2025). There exists a notable reluctance among some individuals to engage with chatbots, perceiving AI-driven chatbots as deficient in knowledge and empathy (Ruan & Mezei, 2022). Approximately 60% of customers still express a preference for conversing with human agents due to reasonable wait times and the humanized services that display compassionate care (Chi et al., 2021; Xie et al., 2023). This divergence between rapid organizational adoption and persistent consumer skepticism underscores a pressing practical challenge: firms must learn how to mitigate users' negative perceptions and sustain their intention to use

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AI-driven chatbots (Lu & Zhang, 2025). With chatbot users exceeding one billion, understanding the determinants of continuous use intention has become a pressing agenda for converting initial deployment into sustained usage and tangible business value (Niu & Mvondo, 2024; Rese et al., 2020).

Some scholars contend that the continuous use intention constitutes a pivotal element in comprehending the utilization of AI-driven chatbots (Nan et al., 2025; Thi & Duong, 2025). Continuous use intention pertains to the depth of an individual's foresight regarding their ongoing engagement with digital services in the forthcoming period (Bhattacharjee, 2001; Molinillo et al., 2023; Shiao & Luo, 2013), which signifies the projected future utilization or consumption, closely intertwined with actual usages (Molinillo et al., 2023). While prior work has examined continuous use intention across diverse technologies, three gaps remain salient in the context of AI chatbots. Firstly, the majority of studies pertaining to continuous use intention have centered in the realm of conventional media and network communication technologies (Shiao & Luo, 2013; Singh et al., 2021). However, existing studies concerning AI use intention also focus more on initial intention rather than continuous intention (Nan et al., 2025; Thi & Duong, 2025). AI-driven chatbots differ from conventional communication technologies in that they are conversational, function similarly to autonomous agents, and possess generative capabilities, enabling users to co-produce content and decisions through iterative interaction. These characteristics may reshape post-adoption evaluations and make continuous use depend on both usefulness and users' perceptions of novelty (Hwang et al., 2019; Nan et al., 2025; Pang et al., 2025). Therefore, examining continuous use intention in chatbot contexts facilitates the test of boundary conditions within existing theoretical frameworks and could clarify whether established mechanisms hold in AI-mediated interactions.

Secondly, prior research on novel technology use has predominantly relied on adoption-oriented frameworks such as the technology acceptance model (TAM), the theory of planned behavior (TPB), and the unified theory of acceptance and use of technology (UTAUT) to explain users' behavioral intention toward emerging services (Alalwan et al., 2017; Patil et al., 2020). Meanwhile, continuous use intention has also been extensively examined through the expectation confirmation model (ECM) (Nan et al., 2025; Orden-Mejía et al., 2025), which emphasizes post-adoption evaluations such as confirmation and satisfaction. Despite their prominence, these frameworks each provide only a partial account of continuous use intention toward AI-driven chatbot services. Adoption-oriented models primarily foreground instrumental beliefs and utility expectations (e.g., effortlessness anticipation and performance anticipation), whereas sustained chatbot engagement additionally relies heavily on experiential gratifications (e.g., psychological gratification, interactive gratification, and monetary gratification) that motivate repeated interaction. In contrast, ECM centers on the sequential appraisal of confirmation and satisfaction (Bhattacharjee, 2001; Oliver, 1980), which may not fully capture the dual motivations underlying continued engagement with AI-driven chatbots. To address this gap, this study draws on behavioral reasoning theory (BRT) as an integrative framework, integrating UTAUT2 and uses and gratifications theory (U&G) to theorize instrumental anticipations and experiential gratifications as complementary reasons to probe the pathways jointly shape continuous use intention toward AI-driven chatbots (Alkawsi et al., 2021; Cheng & Jiang, 2020; Westaby, 2005).

Thirdly, the majority of previous investigations have predominantly examined the inherent traits of AI-driven chatbots, including quality of service, anthropomorphism, and responsiveness, to elucidate their influence on users' behavior (Ashfaq et al., 2025; Hsu & Lin, 2023; Sun et al., 2024). Nonetheless, feature-centric explanations are informative but insufficient for guiding retention, because continuous use intention ultimately depends on how users interpret these features and whether they are willing to invest effort in sustained interaction (Orden-Mejía et al., 2025; Thi & Duong, 2025). In algorithm-driven services, as users

recurrently update both the perceived value of the service and the justification for continued effort (Hong et al., 2025; Sharma et al., 2025), continuous use intention could be shaped by cognitive appraisals that evolve through repeated experiences (Ashfaq et al., 2025). Accordingly, two proximal experiential mechanisms become central: perceived innovativeness, which reflects users' novelty assessment of the chatbot as an innovative technology, and active involvement, which captures sustained investment in the interaction as a form of user engagement. This underscores the need to model perceived innovativeness and active involvement as mediating drivers of continuous use intention. Addressing this gap can specify how user-level reasons translate into continuous use intention, and it offers actionable implications for chatbot design and user engagement strategies.

Inspired by these identified lacunas and grounded in an integrated paradigm, this study proposes a conceptual model to elucidate the mechanisms underlying continuous use intention toward AI-driven chatbots (Fig. 1). Building on BRT as an overarching lens and incorporating complementary insights from UTAUT2 and U&G, the model links users' anticipations and gratifications to continuous use intention through perceived innovativeness and active involvement. Accordingly, the study pursues the following research objectives: (1) to theorize and test how users' anticipations of chatbot use, namely effortlessness anticipation and performance anticipation, inform their appraisal of perceived innovativeness; (2) to explicate how gratification-based motivations, including psychological, interactive, and monetary gratifications, translate into sustained use intention toward AI chatbots through active involvement; (3) to clarify the mechanisms through which perceived innovativeness and active involvement shape continuous use intention toward AI-driven chatbots.

By addressing these objectives, this research markedly propels the current body of quantitative academic analysis on AI-driven chatbots with multifaceted contributions. Firstly, it enriches the contemporary research context concerning user behavior, particularly regarding continuous use intention, by exploring key dimensions of users' perceptions regarding AI-driven chatbot utilization. Secondly, by formulating an integrated theoretical framework, this article extends the explanatory power of established theoretical frameworks of BRT, UTAUT, and U&G within the burgeoning technological domain of AI to elucidate the influences of chatbots on users. Thirdly, this study broadens prior literature concerning user behavior by exemplifying the mediating roles of perceived innovativeness and active involvement in shaping users' continuous use intention toward AI-driven chatbots. Moreover, this study also furnishes stakeholders with insights into enhancing their sustainability by comprehending the attitudes and behaviors of chatbot users, alongside implementing strategies to ameliorate user experience and augment user engagement.

2. Literature review

2.1. Theoretical foundation

BRT, as elucidated by Westaby (2005), posits that consumers' reasons for or against a behavior constitute antecedents to their motivation to engage in specific actions. These reasons are delineated as specific cognitive constructs linked to behavioral explications. Consumers' awareness of benefits (rationales for) and risks (rationales against) informs their reasoning process and decision-making, thereby influencing their perceptions of value and usage intention of specific products or services (Molinillo et al., 2023). BRT has increasingly been adopted to explain user responses to AI services, including AI chatbots (Ashfaq et al., 2025) and conversational agents (Jan et al., 2023). These studies have considered the impact of various characteristics of AI on consumer trust, stickiness, affinity, and willingness to use through multiple conceptual lenses. Given these insights, the present study advocates for the utility of BRT as a robust framework for comprehending the facilitators and obstacles influencing continuous use intention toward AI-driven

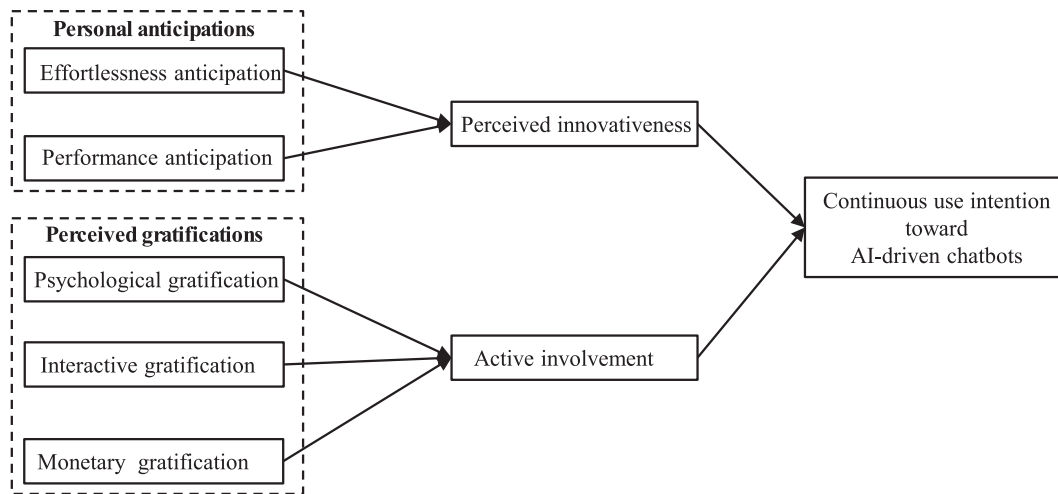


Fig. 1. Conceptual model.

chatbots. Since the variability of these reasons may vary across different technological contexts, the integration of BRT with other theoretical frameworks (e.g., UTAUT and U&G) can enhance the explanatory capacity of behavioral models in an AI-driven context (Patil et al., 2020; Sadiq et al., 2025; Xie et al., 2023).

Venkatesh et al. (2003) crafted the comprehensive UTAUT by systematically reviewing eight prominent technology adoption models. According to UTAUT, behavioral intention is directly influenced by social influence, effort expectancy, performance expectancy, and facilitating conditions (Patil et al., 2020). Afterward, UTAUT2 was introduced (Venkatesh et al., 2012), which is an expanded iteration of UTAUT customized for the sphere of consumer technology acceptance. Venkatesh et al. (2012) recommended that their model be examined in different national contexts, applied to a wider range of innovative technologies, and extended by incorporating additional relevant factors in consumer settings. UTAUT2 enhanced the explanation of variation in behavioral intention and actual technology use (Sadiq et al., 2025). Since its inception, UTAUT2 has garnered widespread acceptance and application in various contexts, particularly in consumption scenarios due to the proliferation of technology services, applications, and platforms targeting consumers (Patil et al., 2020; Venkatesh et al., 2012). UTAUT2 has also emerged as a pivotal framework for examining the adoption of AI technologies across diverse application domains, including AI-generated content in design (Li, 2025), chatbots and recommendation engines in social commerce (Li, 2025), and ChatGPT in education (Foroughi et al., 2024), with a particular focus on users' behavioral intention. Given these considerations, UTAUT2 was chosen as one part of the theoretical foundation for the investigation, wherein the mediating function of perceived innovativeness between personal anticipations and consumers' continuous use intention toward AI-driven chatbots was examined.

Originating from studies in radio communication, U&G has gradually transformed into a fundamental framework employed to comprehend the rationale and sociopsychological motive behind individuals' media utilization and the gratifications derived from active media involvement, with a user-centric perspective (Katz, 1974). U&G posits that consumers exhibit proactivity, selectivity, motivation, and preference in their utilization of various media platforms (Katz, 1974). This framework has been extensively employed in traditional media studies, including television, internet websites, and cellphone investigations (Cheng & Jiang, 2020; Gan & Li, 2018; Xie et al., 2024). With the proliferation of novel media technologies, researchers begin to integrate U&G into emerging platforms such as MySpace, podcasts, Facebook, and Twitter, which have outlined the intricate user motivations for utilizing online digital services, including hedonic, psychological, social,

interactive, utilitarian, technological, effortless, and monetary gratifications (Cheng & Jiang, 2020; Gan & Li, 2018; Xie et al., 2024). It has also been harnessed to examine users' behaviors (e.g., purchase intention, continuous use intention) of AI-driven services, such as AI voice-based assistants, online shopping e-services, a text-based retailing chatbot named *Emma*, and AI chatbots (Chung et al., 2020; Kim et al., 2024; Xie et al., 2024). These studies have revealed that users turn to AI services not only for utilitarian purposes, such as acquiring information and solving problems, but also for hedonic pursuits, including seeking emotional support and entertainment (Huang et al., 2025; Kim et al., 2024). Correspondingly, U&G offers a complementary analytical lens for unpacking the multifaceted psychological and social gratifications underpinning AI use. Drawing upon the relevant constructs of U&G, this research centered on the influence of psychological, interactive, and monetary gratifications on individuals' active involvement in AI-driven chatbots.

2.2. Continuous use intention toward AI-driven chatbots

The conceptualization of an AI-driven chatbot can be delineated as an automated conversational system that facilitates interaction with humans or as an artificial entity engineered to engage in dialogue with humans utilizing natural language processing (NLP) as both input and output mediums (Ashfaq et al., 2020; Xie et al., 2024; Zhu et al., 2022). Operating akin to personal secretaries, chatbots offer support to users with ample functions, encompassing information retrieval, search inquiries, and the cultivation of sociable bonds (Ashfaq et al., 2020; Chung et al., 2020). In comparison to conventional search engines and information hotlines, an AI-driven chatbot offers enhanced anonymity and swifter responses by providing multifarious social cues, spanning verbal, visual, auditory, and imperceptible realms (Cheng & Jiang, 2020). Companies deploy these artificial virtual agents to meet customer demands while delivering recreational and informational value (Chung et al., 2020; Xie et al., 2024). The provision of e-service through AI-driven chatbot assistance not only facilitates customer involvement but also delivers interactive customer support. Furthermore, the utilization of chatbot e-services significantly contributes to augmenting customer contentment with relevant brands (Chung et al., 2020). These factors collectively motivate companies to develop strategies to sustain users' continuous use intention toward AI-driven services (Affandi et al., 2025; Nan et al., 2025; Orden-Mejía et al., 2025).

Use intention refers to an individual's mental disposition toward accepting and engaging with a particular technology or service (Davis, 1989), which serves as a reliable predictor of subsequent technology-related usage behaviors (Thi & Duong, 2025). Continuous use

intention of technology indicates users' future use plans, which are tightly linked with their actual usage, playing a pivotal role in comprehending information system (IS) utilization (Ashfaq et al., 2020; Bhattacharjee, 2001; Pathak et al., 2025). As an outcome factor, continuous use intention of a service is a subjective evaluation of an individual's willingness to purchase or use the service based on their previous usage experience (Molinillo et al., 2023), pertaining to the degree of determination that someone possesses to engage in a specific behavior (Molinillo et al., 2023; Singh et al., 2021). Compared to users with negative experiences, individuals who showcase gratifications on applications are prone to generate a strong inclination to persist in utilizing them instead of switching to surrogates (Bhattacharjee, 2001).

With AI rapidly permeating technology-driven services, scholars have increasingly explored what motivates people to engage with generative AI applications. However, much of the existing literature has concentrated on initial adoption intention of AI, leaving the continuous use intention relatively underexamined (Thi & Duong, 2025). For example, focusing on design practitioners, Li (2025) found that performance expectancy, effort expectancy, social influence, and facilitating conditions significantly increase intention to use AI-powered design services, whereas anxiety and perceived risk act as deterrents. Beyond these utilitarian factors, based on a cybernetic lens, Pathak et al. (2025) underscored that informational interaction and emotional alignment are pivotal in reinforcing users' intention to rely on chatbot systems. Notably, an emerging strand of research has begun to shift attention toward continuous intention in AI-based services. For instance, building on the expectation-confirmation model and incorporating complementary theoretical perspectives, Nan et al. (2025) further revealed that continuous use intention toward AI is jointly determined by information and system quality, privacy concerns, and perceived innovativeness. Within the AI chatbot domain, Affandi et al. (2025) employed a dual-process framework and showed that users become more engaged when they recognize clear functional benefits from AI assistants, and that emotional expression during interaction can further reinforce intentions to reuse these systems. Likewise, Orden-Mejía et al. (2025) found that destination chatbots generate higher user satisfaction when they provide high-quality information and are perceived as enjoyable and useful; importantly, satisfaction subsequently emerges as a strong driver of continued chatbot usage.

Overall, prior research has helped establish continuous use intention as a central outcome in studies of AI-enabled services. Nevertheless, most evidence has been dominated by a feature-driven, utility-based perspective, largely emphasizing users' expectations about AI functionalities. Although a limited number of studies have acknowledged the relevance of attitude and satisfaction (Nan et al., 2025; Pathak et al., 2025), existing research still offers insufficient insight into users' psychological gratifications and how these internal drivers translate into more active and sustained participation in human-AI chatbot interactions. To bridge this gap, the present study retains key functional expectancy factors while also setting perceived gratification as a core psychological mechanism supporting continuous use intention. Specifically, this study conceptualizes gratification as a multidimensional construct comprising three dimensions, thereby enabling a more nuanced examination of the boundary conditions underlying continuous use intention.

2.3. Personal anticipation

Based on UTAUT2, personal anticipations can be divided into two categories: effortlessness anticipation and performance anticipation (Hateftabar, 2023; Singh et al., 2021; Venkatesh et al., 2012). Effortlessness anticipation has been identified as the ease of utilizing technology as a crucial component in behavioral intention of accepting and employing new technologies (Patil et al., 2020), suggesting that users' willingness to embrace new technologies depends largely on the perceived ease of adoption and the level of effort required (Hateftabar,

2023). Earlier researches have depicted that when a technology is perceived as user-friendly or requires less effort, it tends to reinforce the usage inclination of users (Hateftabar, 2023; Patil et al., 2020; Singh et al., 2021). Within the domain of AI, the effortlessness anticipation hinges upon information quality and waiting time, thus lowering cognitive efforts for users (Kumar et al., 2024; Li, 2025). AI-driven chatbots exhibit a propensity for furnishing comprehensive, timely, impartial, meticulous, and detailed product information through mechanized or analytical learning systems, thereby epitomizing high-quality information (Ruan & Mezei, 2022). Within the paradigm of AI-driven human-computer interaction, the reduction of waiting time can ameliorate a customer's discontent during wait intervals, consequently fostering a marked enhancement in overall satisfaction (Osei-Frimpong et al., 2022; Ruan & Mezei, 2022). Hence, a customer's appraisal regarding information quality and waiting time tied to the responsiveness to their requirement will influence their effortlessness anticipation in the field of AI, thereby exerting a consequential impact on their continuous use intention (Li, 2025; Ruan & Mezei, 2022).

Performance anticipation delineates individuals' perceptions concerning the extent to which the utilization of technologies enhances their job performance (Alalwan et al., 2017; Molinillo et al., 2023; Singh et al., 2021). It characterizes individuals' expectations on the benefits which a technology can yield, such as usefulness, effectiveness, and productivity, in the execution of specific tasks (Patil et al., 2020; Venkatesh et al., 2012). There is a prevailing consensus that when users perceive potential benefits stemming from the adoption of novel technology, their propensity to embrace and employ it will dramatically gravitate higher (Alalwan et al., 2017; Hsu & Lin, 2023; Patil et al., 2020). Prior scholarship has extended the notion of service performance to AI services (e.g., AI-driven chatbots), with empirical attention directed toward the linkage between AI performance and users' behavioral responses (Niu & Mvondo, 2024). Through advanced data processing and algorithmic capabilities, AI systems are able to address personalized and complex user demands (Guha et al., 2023). DeLone and McLean (2003) proposed an evaluative framework for technical performance expectations, emphasizing criteria such as compatibility, accuracy, and responsiveness, which have been adopted in subsequent assessments of AI performance (Affandi et al., 2025). Existing evidence further indicates that information accuracy and perceived trustworthiness strengthen users' reliance on AI assistants and foster favorable attitudes, thereby encouraging continued use when users encounter recurring needs or problems (Kumar et al., 2024). Responsiveness, as a critical driver of timely and precise service delivery, has been shown to elevate overall evaluations of the utility of AI products (Pelau et al., 2021). Compatibility contributes to maintaining consistency across user experience and technological functioning and is often regarded as the dimension most strongly associated with personalization-related perceptions that facilitate the adoption of contemporary AI systems (Affandi et al., 2025). Previous studies have examined how AI performance impacts user satisfaction and loyalty, which highlighted performance anticipation as an influential factor in shaping users' beliefs and intentions about the adoption of AI technology (Hateftabar, 2023; Hsu & Lin, 2023; Pathak et al., 2025).

2.4. Perceived innovativeness

Innovativeness denotes an individual's tendency to actively pursue novel and distinct experiences (Hateftabar, 2023; Patil et al., 2020). Perceived innovativeness captures the extent to which users judge a given product or service to embody meaningful innovative qualities in the marketplace, most notably novelty and uniqueness (Nan et al., 2025). The manner in which an individual demonstrates receptiveness toward new experiences or willingness to explore intriguing possibilities can be interpreted as an indication of their perceived propensity toward innovativeness (Molinillo et al., 2023). From a market perspective, innovation has long been recognized as a key driver through which firms

and brands secure competitive differentiation and strategic advantage (Nan et al., 2025). At the user level, perceived innovativeness has been endorsed as an essential indicator of the adoption of new products or innovations in multiple fields (Molinillo et al., 2023; Patil et al., 2020). Perceived innovativeness can be regarded as the manifestation of a person's tendency to take risks within the field of innovative technologies, which is present in some persons but absent in others (Molinillo et al., 2023; Moussawi et al., 2023). Variations in individual personality give rise to diverse attitudes and behaviors of innovativeness, leading to a dichotomy in consumer behavior (Patil et al., 2020). Although some users exhibit a conservative approach and display hesitancy in adopting new technologies, others assume the role of pioneers in embracing and implementing innovative solutions (Alkawsi et al., 2021). Empirical evidence further suggests that perceived innovativeness meaningfully influences users' evaluative experiences and subsequent behavioral intentions. For instance, Hwang et al. (2019) suggested that users are more likely to form favorable evaluations of a technology when it is perceived as innovative and unique. Extending this line of inquiry to the context of AI-based services, Nan et al. (2025) demonstrated that perceived innovativeness enhances users' satisfaction with AI services, thereby contributing to their continuous use intention toward such services.

2.5. Perceived gratification

Synthesizing the previous literature of U&G, we distinguish three dimensions of perceived gratification: psychological, interactive, and monetary gratifications (Hewei, 2022; Osei-Frimpong et al., 2022; Singh et al., 2021). Psychological gratification signifies the efficacy that arises from the emotions evoked by a certain application or product (Osei-Frimpong et al., 2022). The experience of psychological gratification can be achieved when technical services (a product or service) elicit users' excitement, pleasure, and enjoyment, which corresponds to users' behavioral intention to engage in a service (Osei-Frimpong et al., 2022; Singh et al., 2021; Xie et al., 2023). Within human-machine interaction settings, chatbots are increasingly positioned as both conversational counterparts and affective support agents (Huang et al., 2025). Prior research indicates that engagement with chatbots can enhance individuals' psychological well-being (Pham et al., 2022), insofar as personalized and interactive services enable these systems to satisfy users' emotional needs (Kim et al., 2024). Additionally, loneliness is often accompanied by perceived deficits in social connectedness, prompting individuals to compensate by strengthening affiliative ties with non-human entities such as AI chatbots (Peng et al., 2025). Hence, humans will seek intimacy from human-AI interaction, which is a psychosocial procedure featuring communication and emotions that involves trust (Chiang et al., 2022; Xie et al., 2023). Strong human-computer intimacy could enhance psychosocial gratification, which can be ascribed to the fundamental human nature of longing for intimacy and connectedness (Chiang et al., 2022). A high level of intimacy indicates that humans recognize AI-driven chatbots to be knowledgeable and courteous with the conviction that chatbots are capable of immediately addressing their concerns (Ashfaq et al., 2020). These positive assessments will bolster their trust in the AI-driven services and heighten psychological gratification (Ashfaq et al., 2020; Chiang et al., 2022).

Interactive gratification pertains to the enjoyable sense of completion that arises from assessing how effectively a service or product fits specific needs or objectives (Xie et al., 2023). The interactive content can be categorized into two main types: social interaction and task-based interaction (Hatefbar, 2023). Previous IS research has investigated several aspects of interactions, including controllability, responsiveness, communication, associativity, and personalization, which have influences on users' behavior (Osei-Frimpong et al., 2022). Users' interaction via the utilization of digital services stipulates their leaning toward seeking diversity and attaining gratification, which has gained prominence in the study of user behavior within the AI arena (Ashfaq et al., 2020). Extant scholarship on chatbot use indicates that

conversational engagement is facilitated by chatbots' affiliative interactional cues, as these systems can instantiate a relatively egalitarian and non-evaluative dialogic space compared with human interlocutors (Huang et al., 2025). Such a communicative setting can attenuate users' fear of negative evaluation (Ali et al., 2024). The perceived absence of negative judgment has consequently been positioned as a pivotal antecedent of human-computer relationship development; as users experience enjoyment during chatbot interaction, affective attachment and satisfaction are likely to arise in tandem (Kim et al., 2024, 2025). In parallel, the "non-human" status of chatbots may diffuse the social pressures intrinsic to human communication, thereby reducing social anxiety and amplifying interactive gratification (Huang et al., 2025).

Monetary gratification refers to the perceived value that is capable of being quantified by evaluating the economic benefits and costs tied to an exchange or the usage of services and products (Hewei, 2022; Nicolau et al., 2020; Venkatesh et al., 2012). Consistent with the principles of equity theory, individuals experience gratification when they sense a fair ratio between the outcomes they receive and the inputs they contribute, which indicates that service performance is directly proportional to the monetary value attributed to the service. In contrast, when the individual's benefit is lower than the resources invested, a state of non-gratification arises (Nicolau et al., 2020). The definition of monetary gratification elucidates the superior financial benefit in relation to alternative choices (Nicolau et al., 2020). Perceived monetary costs, encompassing monetary or non-monetary expenditures including time, search effort, and psychological costs, can impede the acceptance of innovative technologies (Hong et al., 2025; Sharma et al., 2025). The provision of convenient services with high monetary value can significantly influence perceived gratification (Hewei, 2022). In the context of AI chatbots, monetary gratification is contingent upon customers' cost-benefit appraisal in AI dialogic communication. The AI dialogic attributes, characterized by responsiveness and conversational finesse, can emulate human interaction, thereby fostering an easy and familiar experience for customers. Their comments, feedback, grievances, and requirements are handled promptly, making them perceive their utilization of chatbots as a rewarding experience (Xie et al., 2023). Sharma et al. (2025) reported that users tend to evaluate AI services more favorably when monetary costs are viewed as reasonable and search performance is enhanced, which in turn strengthens adoption intention.

2.6. Active involvement

Users' involvement refers to the active participation of users in mutually beneficial relationships with a corporation (Osei-Frimpong et al., 2022; Shiao & Luo, 2013). Following the proposal of involvement, this concept was subsequently employed by Alalwan et al. (2017) in online digital services, where it is regarded as a key component of user behavior. Involvements can be split into two distinct categories: passive and active (Kang et al., 2014). Passive individuals in an online community are only involved in browsing the content and avail themselves of various advantages offered, yet refrain from actively participating in community interaction (Osei-Frimpong et al., 2022). It has been highlighted that passive individuals who observe but do not actively participate in an online communal interaction may not significantly promote its overall effectiveness and achievements (Casaló et al., 2010; Kang et al., 2014; Shiao & Luo, 2013). Active involvement can also contribute to the improvement of individuals' comprehension of a brand, thereby enabling them to assist other members in resolving issues related to a product or service usage and making well-informed choices of purchasing or continuing usage (Casaló et al., 2010). In AI contexts, involvement is commonly conceptualized as an individual's subjective assessment of personal relevance toward AI services, shaped by internal needs, values, and interests, and users' active involvement with chatbots refers to their proactive and sustained engagement with such systems (Pang et al., 2025). Prior research has underscored active involvement as a key factor in fostering attitudinal loyalty and identification,

particularly in the context of chatbot-based services (Gomes et al., 2025; Shahzad et al., 2024).

3. Hypotheses development

3.1. Linking personal anticipation to perceived innovativeness

Past research discovered that the factors of effortlessness anticipation and performance anticipation have notable impacts on perceived innovativeness (Alkaws et al., 2021; Patil et al., 2020; Venkatesh et al., 2012). Guided by UTAUT2, various factors, involving recognized hazard, perceived enjoyment, perceived innovativeness, perceived value, effortlessness anticipation, and performance anticipation have been thoroughly investigated by multifarious academics, and have been utilized in AI-related investigations (Li, 2025; Sadiq et al., 2025). Among these factors, effortlessness and performance anticipation have been uncovered as crucial variables in several studies that assessed the utility and accessibility of AI technologies and have positive influences on users' perceived innovativeness, enabling them to accept a new service at an earlier stage (Li, 2025; Pathak et al., 2025).

Within the arena of online internet services, users emphasize convenience and adaptability when it comes to controlling and retrieving content and information (Patil et al., 2020). Numerous online services deliver characteristics such as user-friendliness, entertainment value, and personalization, which contribute to the improvement of their effortlessness experience (Hsu & Lin, 2023). Effortlessness anticipation also plays a pivotal role in shaping how individuals engage with AI-enabled assistive technologies. Existing evidence suggests that perceived ease of use, particularly factors such as system usability and the steepness of the learning curve, substantially determines users' motivation to learn and subsequently adopt AI technology (Li, 2025). When an AI tool features a relatively gentle learning curve, users can master it more easily and are more likely to perceive the technology as innovative (Li, 2025; Singh et al., 2021). Hence, the article assumed:

H1. Effortlessness anticipation is correlated with perceived innovativeness positively.

Advanced AI capabilities such as affective analytics, computational data processing, speech and audio recognition, and visual understanding are increasingly integrated into chatbot systems to strengthen decision support and elevate interactional efficiency (Pathak et al., 2025). As these technologies mature, chatbots have gradually moved beyond rule-based exchanges and routine information delivery. Contemporary systems are increasingly capable of modeling conversational regularities in human communication and producing contextually attuned solutions for a broad spectrum of user concerns (Cheng & Jiang, 2020; Chiang et al., 2022). Such performance advancements improve both the speed and competence with which chatbots respond to user demands, thereby increasing the likelihood of eliciting awe, a heightened emotional state that arises when service experiences substantially exceed users' initial performance expectations (Baird & Maruping, 2021). When perceived performance reaches or surpasses what users anticipate, their acceptance of and openness to innovativeness is correspondingly strengthened (Alkaws et al., 2021; Pathak et al., 2025). Thereby, this research formulated the following hypothesis:

H2. Performance anticipation is correlated with perceived innovativeness positively.

3.2. Linking perceived gratification to active involvement

Numerous online studies, spanning diverse management domains such as hospitality, tourism, marketing, social media platforms, streaming platforms, and AI-driven services, have examined the phenomenon of involvement, which fluctuates by a range of factors (Hewei, 2022; Huang et al., 2025; Nicolau et al., 2020). Scholars are increasingly

exploring alterations in customer experience and involvement resulting from AI-driven chatbots, as media technologies integrate AI to augment communication efficiency (Ruan & Mezei, 2022). These factors encompass psychological benefits, interactive gratification, and monetary value in AI contexts (Huang et al., 2025; Kim et al., 2024; Sharma et al., 2025).

Regarding psychological gratification, U&G theory, which originated in the documents of social psychology, has acquired prominence in the research of online networks and computer-mediated communication (Gan & Li, 2018; Osei-Frimpong et al., 2022). It posits that individuals' choices of online consumption are determined by the capacity of certain elements within the online services to gratify their psychological and social requirements (Kim et al., 2024). This framework asserts that the social environment proposes many incentives for psychological gratification, facilitated by the interaction between specific fundamental necessities and an individual's unique qualities (Osei-Frimpong et al., 2022). U&G highlights that users will show an elevated degree of involvement in using digital services if they gain psychological gratification (Gan & Li, 2018). Rese et al. (2020) unveiled that the adoption of *Emma*, a chatbot utilized by retailers of fashion for assisting with pre-purchase inquiries, could be amplified through various gratifications. Cheng and Jiang (2020) ascertained that users' gratifications serve as catalysts for their involvement with mental health chatbots through an empirical investigation. Recent evidence suggests that AI chatbots may serve as emotionally supportive interlocutors, enabling individuals to disclose personal thoughts and regulate distressing affective states (Huang et al., 2025). Huang et al. (2025) further demonstrated that chatbot-mediated interactions can partially compensate for unmet emotional needs in offline interpersonal communication, thereby fostering more positive affect and encouraging deeper, more proactive user participation. Hence, this article proposed:

H3. Psychological gratification is correlated with active involvement positively.

Several social theories, involving social exchange theory (SET) and social identity theory (SIT), explain how individuals' interactive gratification escalates their incentive for online community involvement (Casaló et al., 2010; Meng & Leung, 2021). SET alludes to the interaction between individuals, in which they anticipate receiving social benefits such as approbation, status, and respect by being involved in a community (Meng & Leung, 2021). While there is no assurance of receiving benefits, individuals who are part of an online community are inclined to make contributions and anticipate reciprocal interactions, assistance, and valuable information from others (Shiau & Luo, 2013). SET suggests that increased levels of member interaction within a community will foster greater involvement in communal activities for individuals (Meng & Leung, 2021). SIT states that individuals who perceive potential advantages from pursuing social interactions within a particular community are more likely to develop a social identity related to the community, thereby undergoing a sense of affiliation and inclusion (Casaló et al., 2010; Kang et al., 2014; Meng & Leung, 2021). Those who self-identify as part of a virtual community are more likely to embrace the community and be actively involved in its activities (Kang et al., 2014). Related evidence has likewise been documented in AI chatbot settings, where chatbot-mediated interaction tends to lessen users' concerns about being judged by others (Ali et al., 2024). By mitigating fear of negative evaluation and easing social anxiety, chatbots can further relieve the interpersonal pressures that often characterize human-to-human communication, thereby facilitating stronger user involvement in AI-driven chatbot engagement (Huang et al., 2025; Kim et al., 2024, 2025). Based on these arguments, this paper assumed:

H4. Interactive gratification is correlated with active involvement positively.

For monetary gratification, in the realm of tourism, there is a notable and favorable correlation between monetary gratification and

behavioral intentions, including the likelihood of advocating the tour to others and involvement in future tours (Nicolau et al., 2020). Furthermore, it's reported that behavioral intentions to engage in future involvement were affected by perceived gratifications, which included functional, monetary, emotional, social, and novelty perspectives (Singh et al., 2021). These gratifications influence future travel behavior and are closely linked to the projected advantage or profit derived from earlier experiences of consuming group trips (Nicolau et al., 2020). In recent times, online services have increasingly been assigned monetary and convenient gratifications (Nicolau et al., 2020; Singh et al., 2021). Several studies of internet-based platforms and AI-driven services have emphasized the significance of monetary gratification in examining individuals' involvement. These studies have verified that people perceive value as an experience and evaluate the cognitive aspects of value obtained via online platforms (Chen, 2021; Osei-Frimpong et al., 2022; Singh et al., 2021). Parallel evidence has also emerged in AI-enabled service contexts, suggesting that monetary considerations are pivotal in shaping users' evaluations of AI technologies. Sharma et al. (2025) demonstrated that in the tourism domain, travelers tend to appraise AI services more favorably when the pricing is viewed as reasonable, and the system delivers superior search performance, which in turn strengthens their adoption intention. Hong et al. (2025) reported that in AI robot services, the heightened monetary burden perceptions undermine perceived value and subsequently suppress users' active involvement. Therefore, the hypothesis was raised:

H5. Monetary gratification is correlated with active involvement positively.

3.3. Linking perceived innovativeness and active involvement to TikTok continuous use intention

Many scholars have reached the consensus that a person's level of innovativeness can have a significant influence on the acceptance of novel technical advancements (Alkawsi et al., 2021; Foroughi et al., 2023; Patil et al., 2020). Individuals with a higher propensity for inventiveness are intended to accept a new technology (Alkawsi et al., 2021). Individuals endowed with a proclivity for inventiveness demonstrate a superior capacity to cultivate positive attitudes compared to their counterparts regarding the prospects of adopting novel technological services (Foroughi et al., 2023). Prior studies have confirmed perceived innovativeness to be a potent psychological and cognitive factor that exerts a substantial influence on behavior intentions of users (Foroughi et al., 2023; Nan et al., 2025; Thi & Duong, 2025). Individuals who actively pursue inventiveness are predisposed to embracing technological advancements and alterations, showcasing a heightened level of innovativeness and deriving gratification from experimenting with novel services (Alkawsi et al., 2021). Perceived innovativeness stands out as a significant determinant of individuals' inclination to continue utilizing a specific service or product (Alkawsi et al., 2021). Perceived innovativeness holds the potential to leave a contrasting impact on continuous use intention during the post-adoption stage (Foroughi et al., 2023; Nan et al., 2025). Accordingly, this study posited:

H6. Perceived innovativeness is correlated with continuous use intention toward AI-driven chatbots positively.

Furthermore, involvement is an indispensable prerequisite for the continuous intention of users (Ashfaq et al., 2020; Kang et al., 2014). The notion of continuous intention to use is regarded as a more suitable variable in technology adoption research, particularly when users exhibit prior familiarity with online involvement technology (Casaló et al., 2010; Lee & Lin, 2023). Individuals who are actively involved in their community will maintain an unwavering intention to continue utilizing the platform (Jiang et al., 2022; Meng & Leung, 2021; Osei-Frimpong et al., 2022). Active users can be stimulated by their involvement in an online service to get a specific affective condition,

such as agreeability, enjoyment, gratification, and amusement (Ashfaq et al., 2020). Those users are inclined to revisit the platform in the future (Shiau & Luo, 2013; Singh et al., 2021). Hence, active involvement in an online platform can potentially improve a user's mental well-being, which subsequently influences their use intention toward the service (Kang et al., 2014; Osei-Frimpong et al., 2022). AI-driven chatbots, distinguished by their ability to evoke social presence and perceived agency, create a distinctive interactional milieu that integrates technological instrumentality with affective engagement (Fang et al., 2025; Gong, 2025). Within such settings, active involvement can amplify relational processes and enrich user-robot interactions, thereby establishing conditions conducive to loyalty and identification (Liu et al., 2024). Furthermore, Pang et al. (2025) demonstrated that active involvement serves a crucial mediating role between perceived value and users' psychological sense of belonging, clarifying how perceived value translates into deeper engagement and continuous long-term interaction. Following these arguments, the paper assumed that:

H7. Active involvement is correlated with continuous use intention toward AI-driven chatbots positively.

4. Research methodology

4.1. Sample and data collection

Given the aim of the present study, prospective participants must possess prior experience with AI-driven chatbot services such as robot assistants, emotional chatbots, intelligent customer service robots, and smart home chatbots. At the beginning of the questionnaire, we provided an operational definition of AI chatbots and introduced several common types, including robot assistants, emotional or companion chatbots, intelligent customer service chatbots, and smart home chatbots, to ensure a shared understanding. Respondents were asked whether they had ever used any of the above AI chatbot services, and those who answered "no" were instructed to exit the survey and were not included in the final dataset. Online data collection, acknowledged as a highly effective method for academics to gather a substantial number of respondents (Xie et al., 2024), facilitated this study from March 19 to April 2, 2024, via an online empirical survey platform managed by Sojump (<http://www.sojump.com>). Sojump is recognized as a premier web-based surveying platform with a substantial user base in China, making it exceptionally suited for online data acquisition across various demographics (Ruan & Mezei, 2022).

Participants were recruited using a convenience sampling approach through both online and offline channels. Specifically, the survey link was posted across multiple social media platforms and online communities, and the researchers also invited potential participants offline by presenting a QR code that linked to the same online questionnaire. Individuals who reported no prior AI chatbot experience or expressed no interest were not invited to proceed. All participants were explicitly informed that their responses would be kept strictly confidential and that no identifiable information would be disclosed to any third parties. Voluntary and anonymous participation was emphasized, allowing participants the freedom to quit from this study at any time. The study protocol was formally approved by the institutional ethics review board at the university of the authors. Ultimately, 613 responses were submitted, with 581 deemed valid after excluding invalid submissions. Invalid submissions were removed if the questionnaire was not fully completed or if the completion time was less than 30 s, which suggested unlikely careful reading and insufficient engagement with the survey. After data screening, 581 valid responses remained for analysis, yielding an effective response rate of 94.8%.

4.2. Measurement

The investigation includes two primary parts. The first part involves

demographic data, years of AI-driven chatbot usage, and the most commonly used chatbot types. The second part encapsulates eight constructs. Drawing upon established literature as the bedrock for assessing progress, the present study benefited from frameworks that were well-developed and widely-acknowledged. Modifications and adaptations were implemented to align with the specific context of AI-driven chatbots. Applying a five-point Likert scale varying from strongly disagree to strongly agree (1–5), each item in this investigation was evaluated. The scales measuring effortlessness anticipation (three items) and performance anticipation (four items) were refined by Venkatesh et al. (2012). The assessment of perceived innovativeness (four items) was adjusted by Singh et al. (2021). The construct of psychological gratification (four items) was derived from Chung and Buhalis (2008). The interactive gratification scale, comprising four items, was revised by Dong et al. (2018). The measurement of monetary gratification (three items) was extrapolated from Singh et al. (2021). The measure of active involvement (four items) was rephrased from Casaló et al. (2010). The continuous use intention scale (four items) was modified upon Bhattacharjee (2001) and Venkatesh et al. (2012). The validity and reliability of all these items have been extensively acknowledged in previous academic research. The detailed measurement items are displayed in Table 1.

4.3. Demographic data

Out of these 581 participants, 316 were determined as male, representing 54.4% of the sample, while the remaining 265 identified as female, constituting 45.6%. Most participants were between the ages of 20 and 49. In terms of educational attainment, 79.9% reported having an undergraduate degree, with 18.4% indicating they hold a master's degree or higher qualification. Furthermore, 59.2% of respondents reported using AI-driven chatbots for more than 1 year. The types of AI-driven chatbots respondents use with high frequency are emotional chatbots, robotic assistants, and intelligent customer service robots, accounting for 37.2%, 28.4%, and 24.8%, respectively. Table 2 presents a comprehensive overview of the demographic characteristics.

5. Data analysis strategy

Initially, the data underwent processing in Microsoft Excel to eliminate invalid responses. Subsequently, primary statistical analyses and the evaluation of common method bias (CMB) were conducted using SPSS 22. Additionally, an examination of the characteristics of AI-driven chatbot users and their usage patterns within the survey sample was performed to glean deeper insights. Following this, Smart PLS 3.2.9 was employed to analyze the models of measurement and structure. The preference for PLS-SEM was grounded on some considerations elucidated. Firstly, PLS-SEM is favored for analyzing intricate models that incorporate multiple intervening factors, like mediators and moderators. Secondly, concerns regarding normality do not arise with PLS-SEM. Thirdly, the utilization of PLS-SEM is more suitable for interpreting data with a smaller quantity of samples. Hence, after thorough deliberation, PLS-SEM was selected for the data analysis (Henseler et al., 2016).

6. Results

6.1. Common method bias (CMB)

This study conducted a principal component analysis employing Harman's one-factor test via the software of SPSS 22, adhering to the methodology established in previous research to analyze CMB. Following Harman's criteria, if a single construction accounts for over 50% of the variation, it indicates the existence of CMB in the data (Ashfaq et al., 2020). The results revealed that the highest variance of a single construct amounted to 30.80%, which is lower than the 50%

Table 1
Measurement of key variables.

Variable	Item	Source
Effortlessness Anticipation (EA)	(1) Learning to use AI-driven chatbots will be easy. (2) AI-driven chatbots will be easy to use. (3) It will be easy to become skillful at AI-driven chatbots.	Venkatesh et al. (2012)
Performance Anticipation (PA)	(1) AI-driven chatbots will be useful in my daily life. (2) AI-driven chatbots will increase my productivity. (3) AI-driven chatbots will provide very quick services. (4) AI-driven chatbots will save more of my time.	Venkatesh et al. (2012)
Perceived Innovativeness (PI)	(1) When I hear about AI-driven chatbots, I look for possibilities to try it. (2) I want to try a new AI-driven chatbots service. (3) I am usually the first to try out AI-driven chatbots services. (4) I like to try new AI-driven chatbots services.	Singh et al. (2021)
Psychological Gratification (PG)	(1) I can seek self-identity on AI-driven chatbots. (2) I get involved with other members on AI-driven chatbots. (3) I can seek a sense of belonging on AI-driven chatbots. (4) I can establish and maintain relationships with other members of AI-driven chatbots.	Chung and Buhalis (2008)
Interactive Gratification (IG)	(1) I am a member of the AI-Chatbot and can communicate freely with others. (2) There are many activities and services about product information, exchange and interpersonal interaction. (3) There are a lot of interaction design features, which can help me achieve the expected goals. (4) AI-Chatbot can give me valuable recommendations and suggestions whenever I need.	Dong et al. (2018)
Monetary Gratification (MG)	(1) I feel that the subscription cost of AI-driven chatbots is not expensive relative to other services. (2) I feel that using AI-driven chatbots offers significant cost savings relative to other services. (3) I feel that using AI-driven chatbots is cost-effective relative to other services.	Singh et al. (2021)
Active Involvement (AI)	(1) I participate actively in AI-driven chatbots activities. (2) I contribute messages, responses other information to AI-driven chatbots. (3) I provide useful information to other members of AI-driven chatbots. (4) I post messages and responses in the community with excitement and on a regular basis.	Casaló et al. (2010)
Continuous Use Intention Toward AI-driven Chatbots (CUI)	(1) I will continue to use AI-driven chatbots if given the opportunity.	Bhattacharjee (2001) and Venkatesh et al. (2012)

(continued on next page)

Table 1 (continued)

Variable	Item	Source
	(2) I will use additional services of AI-driven chatbots introduced in future.	
	(3) I intend to continue using AI-driven chatbots in the future.	
	(4) I will regularly use AI-driven chatbots in the future.	

Table 2Summary of demographic statistics ($N = 581$).

Category	Frequency	%
Gender		
Males	316	54.4
Females	265	45.6
Age		
Under 20 years old	58	10.0
20–29 years old	287	49.4
30–49 years old	196	33.7
Above 50 years old	40	6.9
Education level		
High school and below	10	1.7
Bachelor's degree	464	79.9
Master's degree	97	16.7
Doctor's degree	10	1.7
AI-Chatbots using years		
Below 1 year	237	40.8
1–3 years	288	49.6
Above 3 years	56	9.6
AI-driven Chatbot service most frequently used		
Robot assistants	165	28.4
Emotional chatbots	216	37.2
Intelligent customer service robots	144	24.8
Smart home chatbots	52	9.0
Others	4	0.7

criterion, thus indicating the nonappearance of CMB (Lee & Lin, 2023).

6.2. Measurement model

In line with the guidelines of Hair et al. (2019), the assessment of the measurement model encompasses item loadings, convergent validity, and discriminant validity. All item loadings in the research surpassed the recommended threshold of 0.6, thus affirming their adequacy. The composite reliabilities (CR) fluctuated between 0.804 and 0.850, suggesting satisfactory internal consistency, exceeding the cutoff value of 0.70 (Fornell & Larcker, 1981). Convergent validity was evaluated with the average variance extracted (AVE), with all AVE values surpassing 0.5, validating their acceptability (Fornell & Larcker, 1981). Since all these indices exceeded the suggested benchmarks, the measurement model was considered to exhibit reliable convergence, as displayed in Table 3. Discriminant validity is measured using the square root of the AVE (Table 4), which exceeded the inter-construct correlations for all constructs, signifying their empirical distinctiveness (Wang et al., 2023). Consequently, the examination of discriminant validity was deemed satisfactory. Finally, before analyzing the structural model, an assessment of the entire model was conducted. Model fit can be gauged utilizing the standardized root mean square residual (SRMR) (Henseler et al., 2016). The value of SRMR (0.067) falls below the recommended limit of 0.08, signaling a sufficient model fit. The study also provided the assessment of “Stone-Geisser's Q^2 value”, which gauges the predictive significance of the structure model. The outcomes of blindfolding imply

Table 3

Statistical outcomes of reliability, validity and multicollinearity.

Constructs and items	Loading (>0.6)	t-Value	VIF	CR (>0.7)	AVE (>0.5)
Effortlessness				0.815	0.595
Anticipation (EA)					
EA1	0.732	20.123	1.257		
EA2	0.776	28.976	1.324		
EA3	0.803	34.616	1.287		
Performance Anticipation (PA)				0.835	0.558
PA1	0.788	36.921	1.387		
PA2	0.761	29.576	1.429		
PA3	0.732	22.729	1.393		
PA4	0.705	20.960	1.370		
Perceived Innovativeness (PI)				0.844	0.575
PI1	0.746	33.060	1.377		
PI2	0.751	32.156	1.368		
PI3	0.774	33.285	1.572		
PI4	0.762	31.855	1.480		
Psychological Gratification (PG)				0.847	0.580
PG1	0.755	32.187	1.501		
PG2	0.743	28.543	1.320		
PG3	0.754	31.970	1.482		
PG4	0.794	39.977	1.599		
Interactive Gratification (IG)				0.804	0.507
IG1	0.698	20.003	1.207		
IG2	0.658	15.417	1.261		
IG3	0.741	31.525	1.346		
IG4	0.747	34.251	1.299		
Monetary Gratification (MG)				0.810	0.588
MG1	0.666	14.908	1.243		
MG2	0.813	34.523	1.266		
MG3	0.812	36.186	1.388		
Active Involvement (AI)				0.825	0.542
AI1	0.792	39.802	1.458		
AI2	0.729	29.142	1.260		
AI3	0.694	22.531	1.343		
AI4	0.725	28.331	1.391		
Continuous Use Intention Toward AI-driven Chatbots (CUI)				0.850	0.587
CUI1	0.744	26.399	1.400		
CUI2	0.729	30.162	1.344		
CUI3	0.812	42.133	1.800		
CUI4	0.776	40.697	1.622		

that all Q^2 values of endogenous variables (0.159, 0.211, and 0.338) are above zero, underscoring their remarkable predictive significance (Henseler et al., 2016).

6.3. Multicollinearity

An examination of the variance inflation factor (VIF) was also conducted to ascertain the existence of multicollinearity. It's suggested that a VIF surpassing 5.0 signifies the existence of multicollinearity (Wang et al., 2023). The results of the research revealed that the VIF values (varying from 1.207 to 1.622) for all measured indicators were lower than this threshold, as demonstrated in Table 3, affirming the absence of multicollinearity in this research.

6.4. Structure model

Following the confirmation of sufficient reliability and validity of both the measurement model and the entire model fit, the study employed a bootstrapping technique (bootstrapping subsample = 5000) to examine the presented hypotheses. The path coefficients were computed utilizing Smart PLS, elucidating the relationships among the exogenous and endogenous variables, while also providing the

Table 4

Descriptive statistics and discriminant validity.

	M	SD	EA	PA	PI	PG	IG	MG	AI	CUI
EA	3.931	0.627	0.771							
PA	4.178	0.576	0.445**	0.747						
PI	4.058	0.585	0.417**	0.487**	0.758					
PG	3.711	0.713	0.438**	0.397**	0.413**	0.762				
IG	3.980	0.576	0.430**	0.529**	0.484**	0.590**	0.712			
MG	3.929	0.671	0.380**	0.391**	0.331**	0.396**	0.534**	0.767		
AI	3.721	0.703	0.361**	0.420**	0.535**	0.529**	0.585**	0.427**	0.736	
CUI	4.012	0.599	0.448**	0.521**	0.591**	0.444**	0.595**	0.480**	0.677**	0.766

Notes: Double asterisk (**) represents $p < 0.001$; EA, effortlessness anticipation; PA, performance anticipation; PI, perceived innovativeness; PG, psychological gratification; IG, interactive gratification; MG, monetary gratification; AI, active involvement; CUI, continuous use intention toward AI-driven chatbots; M, the median of latent variables; SD, standard deviation of latent variables; numbers in the diagonal lines represent the square root of the AVE of corresponding constructs.

coefficient of determination values (R^2) for the endogenous constructs of the structural model (Lee & Lin, 2023). The R^2 values for the endogenous constructs were determined to be 28.8%, 40.7%, and 53.2%, respectively, denoting acceptable levels of model overfit. Moreover, all models exhibited significance statistically. A comprehensive overview of the overall findings, encompassing hypothesis testing, path coefficients, t-values, p-values, and results, is delineated in Table 5. The comprehensive model (depicted in Fig. 2) elucidates the key findings. Firstly, the impacts of effortlessness anticipation ($\beta = 0.249$, $t = 6.170$, $p < 0.001$) and performance anticipation ($\beta = 0.378$, $t = 8.887$, $p < 0.001$) on perceived innovativeness were substantiated, statistically supporting the proposed hypotheses of H1 and H2. Secondly, psychological gratification ($\beta = 0.269$, $t = 6.081$, $p < 0.001$), interactive gratification ($\beta = 0.359$, $t = 6.963$, $p < 0.001$), and monetary gratification ($\beta = 0.129$, $t = 2.555$, $p = 0.022 < 0.05$) showed significant influences on active involvement, robustly lending credence to H3, H4, and H5, respectively. Likewise, perceived innovativeness ($\beta = 0.209$, $t = 4.912$, $p < 0.001$) and active involvement ($\beta = 0.378$, $t = 8.268$, $p < 0.001$) exhibit positive influences on continuous use intention toward AI-driven chatbots, corroborating H6 and H7 empirically.

6.5. Mediating effects

This research applied the bootstrap approach to scrutinize the mediating effects. Table 6 delineates the results for both direct and indirect effects observed for every mediating effect. The values of VAF (Variance Accounted For) for each mediating effect were computed to ascertain its mediating effects (Nitzl et al., 2016; Wang et al., 2023). According to Nitzl et al. (2016), a VAF value below 20% implies that there is no mediation, while a VAF value between 20% and 80% suggests partial mediation, and a VAF value above 80% shows full mediation. As shown in Table 6, a partial mediating effect of perceived innovativeness is evident in the relationship between personal anticipations (effortlessness anticipation and performance anticipation) and continuous use intention toward AI-driven chatbots. Similarly, a partial mediating effect of active involvement is observable in the correlations between two kinds of gratifications (interactive gratification and monetary

gratification) and continuous use intention toward AI-driven chatbots. Additionally, it indicates a full mediating effect of active involvement on the relationship between psychological gratification and continuous use intention toward AI-driven chatbots.

7. Discussions and contributions

7.1. Discussions

Drawing from BRT, combined with UTAUT2 and U&G theories, this study endeavors to elaborate on the determinants influencing users' continuous use intention toward AI-driven chatbots. This research presents a referable integrated research paradigm for user behavior research in the AI domain. The findings robustly support all hypotheses posited in the article, confirming the mediation roles of perceived innovativeness and active involvement between these relationships.

Primarily, this study reiterates the prospect of a positive correlation between personal anticipations (effortlessness anticipation and performance anticipation) and perceived innovativeness, aligning with the outcomes delineated by previous research (Alkaws et al., 2021; Kim et al., 2024; Li, 2025; Singh et al., 2021). With the swift advancement of AI, chatbots have emerged as pivotal manifestations of intelligent customer service and personal assistants, progressively evolving to cater to users' anticipations. This evolution enables users to experience technologically-powered innovativeness (Li, 2025; Nan et al., 2025; Pathak et al., 2025). First, AI-driven chatbots leverage NLP to execute speech recognition, semantic comprehension, and sentiment analysis, fostering intelligent interaction. Speech recognition empowers users to engage with chatbots through voice commands, enhancing communication efficiency and convenience. Semantic comprehension facilitates profound analysis of user inputs, enabling chatbots to furnish precise responses and services (Pathak et al., 2025). Sentiment analysis enables chatbots to discern and respond to users' emotional fluctuations, thereby augmenting user experience. Second, users perceive innovativeness through personalized recommendations and tailored services offered by chatbots. By

Table 5

Path results of structural model.

Hypotheses	Path	Standardized β	t-Statistics	p-Value	Results
H1	Effortlessness Anticipation $\rightarrow$ Perceived Innovativeness	0.249	6.170	0.000***	Supported
H2	Performance Anticipation $\rightarrow$ Perceived Innovativeness	0.378	8.887	0.000***	Supported
H3	Psychological Gratification $\rightarrow$ Active Involvement	0.269	6.081	0.000***	Supported
H4	Interactive Gratification $\rightarrow$ Active Involvement	0.359	6.963	0.000***	Supported
H5	Monetary Gratification $\rightarrow$ Active Involvement	0.129	2.555	0.011*	Supported
H6	Perceived Innovativeness $\rightarrow$ Continuous Use Intention Toward AI-driven Chatbots	0.209	4.912	0.000***	Supported
H7	Active Involvement $\rightarrow$ Continuous Use Intention Toward AI-driven Chatbots	0.378	8.268	0.000***	Supported

*** $p < 0.001$.

* $p < 0.05$.

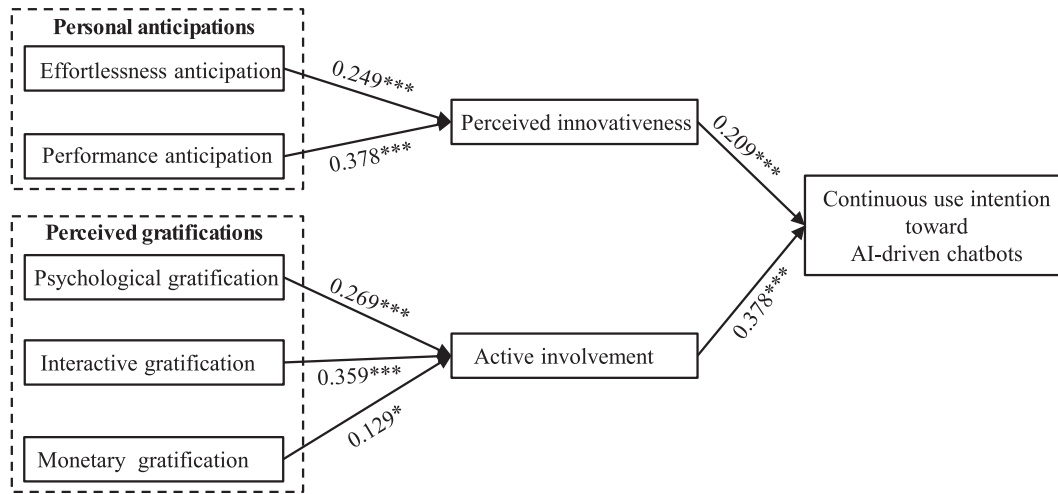


Fig. 2. Path analysis result of the structural model. Note: *** $p < 0.001$, * $p < 0.05$.

Table 6

Mediation testing results.

Path estimated	Direct effect (t value)	Indirect effect (t value)	Total effect	VAF (%)	Interpretation
EA → PI	0.092*	0.052***	0.143	36.36%	Partial mediation
→CUI	(2.442)	(3.513)			
PA → PI → CUI	0.124**	0.078***	0.202	38.61%	Partial mediation
	(3.045)	(4.493)			
PG → AI → CUI	-0.062 ns	0.102***	0.039	261.54%	Full mediation
	(1.471)	(5.458)			
IG → AI → CUI	0.143**	0.137***	0.280	48.93%	Partial mediation
	(3.125)	(5.548)			
MG → AI → CUI	0.113**	0.050*	0.163	30.67%	Partial mediation
	(2.952)	(2.273)			

Note: ns: not significant.

*** $p < 0.001$.

** $p < 0.01$.

* $p < 0.05$.

collecting users' historical interaction data, chatbots curate tailored services such as personalized shopping suggestions, medical advice, and household management, aligning with users' preferences and needs (Gong, 2025). Third, users perceive innovativeness through chatbots' autonomous learning and evolution. Through continual analysis of user queries and interactions, chatbots autonomously refine their responses and interaction modes, rendering more natural and intelligent interactions (Huang et al., 2025; Kim et al., 2024, 2025).

Secondly, this research articulates that perceived gratifications (psychological, interactive, and monetary gratifications) significantly predict active involvement in AI-driven chatbots. These findings are congruent with prior research (Huang et al., 2025; Kim et al., 2024; Sharma et al., 2025). AI-driven emotional chatbots have increasingly adopted a genial and captivating conversational demeanor, leading a growing cohort of individuals to exhibit a willingness to foster friendships with automated chatbots, such as Glow and Replika. In fast-paced daily life, a rising number of individuals are inclined to cultivate an anthropomorphic bond with chatbots, seeking a virtual confidant in the cyberworld (Ali et al., 2024; Huang et al., 2025; Kim et al., 2024, 2025). The amicability, deference, and forbearance exhibited by these chatbots during conversations with users endow psychological value to users and enhance their involvement. AI-driven chatbots employing conversational technology comprise conversational intelligence, social intelligence, and anthropomorphism, which endow chatbots with the

capacity to communicate proficiently and engender a heightened sense of involvement with users throughout dialogues. Chatbots are founded upon language models crafted through deep learning and machine learning methodologies, furnishing a mechanism for constructing a knowledge base across various scenarios, which enables users to effortlessly and economically establish and leverage structured knowledge systems. They also offer a spectrum of services characterized by succinct, lucid, user-friendly functions, mitigating the learning curve for users (Li, 2025). Furthermore, the ubiquity of diverse chatbot variants has led to a reduction in users' costs and amplified their frequency and opportunities to involve in interactions with chatbots.

Thirdly, the affirmative contributions of perceived innovativeness and active involvement to the continuous use intention toward AI-driven chatbots are substantiated, which corroborates the prior outcomes (Ashfaq et al., 2020; Foroughi et al., 2023; Nan et al., 2025; Pang et al., 2025). In light of the attributes delineated by AI-driven chatbots in the antecedent section, the incorporation of nascent technologies (NLP, algorithm recommendation, and deep learning) and the evolution of interactive methodologies (conversational intelligence, social intelligence, and anthropomorphism) augment users' perceived proactivity, connectivity, and communication when involving with chatbots. Simultaneously, the inherent courteous communication style and ethical standards intrinsic to chatbots enable them to exhibit a heightened level of emotional intelligence, thus mitigating potential physical and psychological detriments to users (Huang et al., 2025; Sharma et al., 2025). These factors collectively enhance the user experience, cultivating continuous utilization intent. Moreover, the anthropomorphism of chatbots amplifies user immersion in conversational exchanges, engendering a sense of closeness, while the mechanism of chatbots maintains emotional equilibrium within chatbots, fortifying users' reliance and substantially bolstering their continued usage intention.

7.2. Theoretical and practical contributions

In consideration of these findings, this investigation offers valuable theoretical contributions. Firstly, it broadens the context of research on continuous use intention within the realm of AI-driven chatbots. Previous research on continuous use intention has predominantly centered on conventional media and online communication technologies, like blogs, mobile services, and social media platforms, and existing studies about AI use intention mainly concentrated on users' initial intention, rather than continuous intention. Secondly, the majority of prior investigations have leaned on singular classical research models, such as

TAM, TPB, UTAUT, and ECM, to explore the acceptance of AI technologies. In contrast, by integrating UTAUT2 and U&G within a BRT-based reason framework, this study advances continuous research beyond single-chain post-adoption evaluation models. This research identifies two complementary pathways, perceived innovativeness and active involvement, through which users' personal expectations and perceived gratifications jointly strengthen their intention to continue using the AI chatbot. By articulating these dual mechanisms, the research offers an integrative analytical lens that is applicable to user behavior studies in the AI domain. Thirdly, existing studies have predominantly scrutinized the intrinsic attributes of AI-driven chatbots, such as service quality, anthropomorphism, and responsiveness, to dissect their influences on user behavior. Conversely, this study probes the mechanism of personal anticipations and perceived gratifications as antecedents within the AI-driven chatbot domain from the vantage point of users' subjective experiences, thus proffering a novel perspective on users' technology adoption.

Practically, some managerial contributions merit attention concerning the current investigation. Firstly, developers and designers must prioritize enhancing chatbots' performance to align with users' anticipations, thereby mitigating perceived complexity. This entails refining interaction design and streamlining conversation interfaces, integrating useful features into the dialogue interface, such as emoticons, voice input, and speech recognition, to amplify perceived effortlessness. Continual optimization and rigorous testing are indispensable to enhance the conversational performance, stability, and usability of chatbots. Secondly, developers must concentrate on users' gratifications, considering psychological, interactive, and monetary dimensions. It is a prerequisite for chatbots to possess robust NLP capabilities to discern user intent accurately, which can be achieved by integrating vast corpora into chatbots across various domains like news, social media, and emails. The utilization of sophisticated deep learning methods can bolster chatbots' conversational skills, resulting in smoother, more natural interactions. Designers should apply sentiment analysis technology to chatbots to facilitate a deeper understanding of users' emotional states. Furthermore, advanced intelligent recommendation systems can enhance chatbots' understanding of users' needs, facilitating tailored suggestions and services. When the application of these technologies in chatbots is mature enough, users could perceive reductions in usage costs and learning curves. These strategies stand as the winning formula for attracting and retaining a substantial user base.

8. Limitations and future research directions

It is imperative to highlight certain potential limitations meriting further investigation. Initially, this inquiry commences by prioritizing the perceived perspectives of AI-driven chatbot users, accentuating their usage experiences as pivotal precursors to their behavior. However, the influences of the AI-driven chatbots' traits, such as complexity, integration, responsiveness, dialogism, and utility, on users' perceptions and subsequent behaviors are neglected. Future investigations could comprehensively explore these facets. Secondly, this study does not delve into the research perspective of specific industry sectors. Subsequent research in the domain of AI-driven chatbots could penetrate specific industrial areas like customer service, finance, education, and healthcare to unravel the intricacies of user psychology and behavior after using AI-driven chatbots in these contexts. Lastly, while this study predominantly accentuates the affirmative dimensions of AI-driven chatbots. Future inquiries might scrutinize the ramifications of the attendant risks, such as privacy invasion, leakage, dissemination of false information, and in-depth forgery.

CRediT authorship contribution statement

Yanxinyue Liu: Software, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Kaige Zhang:** Software,

Resources, Project administration, Methodology, Investigation. **Yan Zhang:** Writing – review & editing, Validation, Software, Resources, Investigation. **Hua Pang:** Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Bo Liu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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